Optimal Control of a Wave Energy Converter via Recurrent Reinforcement Learning

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***Abstract*—Despite the potential of wave energy as a sustainable resource, maximizing power capture in a Wave Energy Converter (WEC) remains a challenge, as conventional control strategies often fail to adapt to the inherent irregularity of real-world wave conditions. To address these limitations, this research proposes an AI-driven control architecture designed to maximize energy extraction by adaptively responding to highly irregular and dynamic wave conditions. The proposed architecture utilizes Proximal Policy Optimization (PPO) with an LSTM layer to achieve optimal energy extraction and real-time control efficiency. Experimental results on 200 irregular time-series wave datasets demonstrate that the proposed RL model achieves an average cumulative energy extraction of 3.38×10^8, representing a 8.68% improvement over optimal fixed damping, and it is also comparable or better performance compared to the state-of-the-art controller. These findings suggest that the LSTM-PPO architecture provides a robust and superior alternative for real-time WEC optimization in dynamic marine environments.**

***Index Terms*—Deep reinforcement learning; Energy harvesting; Long short-term memory (LSTM); Optimal control; Proximal policy optimization (PPO); Renewable energy; Wave energy converter (WEC).**

I. Introduction

Ocean wave energy is rapidly emerging as a safe and highly viable renewable energy source [1]. In particular, compared to traditional renewable energies such as solar and wind, ocean waves are relatively more predictable, which facilitates more reliable energy harvesting [2]. Because of these advantages, harnessing wave energy is no longer just a theoretical concept but a practical necessity for meeting future global electricity demands. This is particularly evident in the United States, where the technical generation capacity of marine energy is estimated at a staggering 2,300 TWh/yr [3]. Within this broader category, wave energy alone accounts for a dominant 1,400 TWh/yr, which is an amount equivalent to 34% of the total U.S. electricity generation. To effectively capture this vast and highly dense resource, Wave Energy Converters (WECs) have been established as the primary technology.

While Wave Energy Converters (WECs) are the primary technology for harvesting this power, maximizing energy extraction efficiency remains one of the most significant technical challenges. The primary obstacle derives from the irregular nature of ocean waves [4]. Real sea states are composed of broad-spectrum, non-linear waves with constantly varying frequencies and amplitudes. To cope with this unpredictability, the WEC's internal damping mechanism, known as the Power Take-Off (PTO) system, must be continuously and dynamically adjusted.

To address these environmental irregularities, traditional ocean engineering has widely adopted Model Predictive Control (MPC) as the standard advanced control strategy [5]. MPC attempts to maximize energy extraction by forecasting future wave elevations and solving an optimization problem over a finite horizon. However, the real-world performance of MPC is heavily constrained by its reliance on precise mathematical models of the WEC hydrodynamics. Formulating these analytical models requires domain expertise. Furthermore, to make real-time optimization computationally feasible, these models are typically simplified to linear systems. When subjected to the highly non-linear dynamics of irregular seas, this simplification results in modeling errors that degrade the controller's performance and overall energy efficiency [6].

To overcome the limitations of model-based controllers like MPC, Reinforcement Learning (RL) has emerged as a strong, model-free alternative [7]. RL agents learn optimal control by interacting directly with the environment, bypassing the need for complex mathematical models. Early studies successfully applied deep RL to WECs [8]. However, these previous models lacked the architecture to handle sequential, time-series data. Because ocean waves are continuous and irregular, an agent must interpret past wave patterns to make optimal decisions.

To address this gap, this paper proposes a PPO-LSTM agent to control the RM3 Wave Energy Converter. The LSTM structure enables the agent to effectively process the sequential nature of wave dynamics. In addition, we introduce a custom carry-over system that transfers observation and reward variables between training episodes to maintain system continuity. This approach allows the agent to dynamically maximize Energy Efficiency (EE) while strictly adhering to hard mechanical constraints.

The main contributions of this paper are as follows:

- We propose PPO-LSTM based control strategy that effectively processes the continuous and sequential dynamics of irregular ocean waves to dynamically adjust PTO damping.
- We introduce a custom carry-over logic that transfers observation and reward variables across split training

episodes, ensuring continuous temporal learning without losing context.
- We design a novel EE-based reward function coupled with hard constraints to simultaneously maximize power extraction and guarantee mechanical safety.
- We provide the full implementation of our project as open-source to support future research (https://github.com/oht505/WEC_RL).

II. Background

A. The RM3 Point Absorber

The Reference Model 3 (RM3) is a two-body wave energy converter consisting of a surface float and a submerged spar [9]. As irregular ocean waves induce relative heave (vertical) motion between these two components, a Power Take-Off (PTO) system converts this kinetic energy into electricity. from a control systems perspective, the PTO functions as an adjustable damper. Dynamically tuning this damping coefficient serves as the agent's primary control action to maximize energy extraction from the environment.

However, operating WECs in dynamic oceanic conditions requires strict adherence to hardware safety margins. To prevent mechanical failure, the relative vertical displacement of the float must be rigidly constrained within a limit of $\pm 2.5m$ [9]. Consequently, the primary control objective is to maximize power extraction, subject to these strict mechanical boundaries [4]. Navigating the mechanical boundary within a stochastic wave environment motivates a specialized deep reinforcement learning architecture [8]. Specifically, Proximal Policy Optimization (PPO) is required for its stable policy updates in continuous action spaces, coupled with Long Short-Term Memory (LSTM) networks to effectively interpret time-series data and capture the continuous fluctuations of irregular waves.

B. Proximal Policy Optimization (PPO)

Applying deep reinforcement learning to continuous tasks, such as real-time tuning of the PTO damping coefficient, presents significant stability challenges [10]. Traditional Policy Gradient (PG) methods are sensitive to hyperparameter tuning and step sizes [11]. In a stochastic wave environment, an unconstrained parameter update can cause large policy shifts, degrading control performance. Such performance collapses are particularly significant in WEC applications, as erratic policy behavior translates to violation of the strict mechanical displacement limits. This instability requires an algorithmic approach that rigidly bounds policy updates within a safe operational region.

To overcome the limitations of standard PG methods, PPO is formulated based on an Actor-Critic architecture [12]. Within this framework, the Actor network, denoted as π_θ, observes the current wave state and determines the optimal continuous action, specifically the PTO damping coefficient. Concurrently, the Critic network evaluates the quality of this state by estimating its expected future return, $V(s_t)$. The critical link between these two networks is the Advantage function, defined as:

$$A_t = R_t - V(s_t)$$

In a stochastic wave environment, a high raw reward (R_t) might be the result of a naturally massive incoming wave rather than an optimal control decision. The Advantage function mitigates this ambiguity by subtracting the Critic's baseline expectation from the actual return. This ensures the agent evaluates the true effectiveness of its chosen damping action, independent of the inherent environmental variance [13].

To mathematically restrict the step size of policy updates, PPO tracks the divergence between the current and previous policies using a probability ratio, $r_t(\theta)$, expressed as [10]:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$$

An unconstrained maximization of this ratio can lead to excessively large policy shifts. To ensure stability, PPO utilizes a clipped surrogate objective function [10]:

$$L^{CLIP}(\theta) = \mathbb{E}_t[\min(r_t(\theta)A_t, \text{clip}(\hat{r_t}(\theta), 1-\epsilon, 1+\epsilon)A_t)]$$

where ϵ is a hyperparameter defining the clipping threshold. The min operator establishes a conservative lower bound by penalizing the objective if the policy moves too far from the old one. This clipping mechanism forces the agent to optimize the PTO damping actions within a stable trust region, effectively preventing erratic control updates that could breach the physical safety boundaries of the system.

C. Long Short-Term Memory (LSTM) Networks

Long Short-Term Memory (LSTM) networks are a specialized class of Recurrent Neural Networks (RNNs) designed to model long-term temporal dependencies while mitigating the vanishing gradient problem [14]. Unlike standard feedforward networks that process inputs independently, LSTMs maintain an internal memory mechanism known as the cell state, which functions as a linear pathway to preserve historical information across time steps. The flow of this information is dynamically regulated by three internal gating mechanisms: the forget gate, the input gate, and the output gate. By systemically removing irrelevant past data and incorporating noticeable new inputs, these gates enable the network to maintain a persistent and continuous representation of sequential contexts without numerical instability.

In wave energy conversion control, ocean wave profiles exhibit non-linear and stochastic time-series characteristics. Standard Multi-Layer Perceptrons (MLPs) evaluate inputs instantaneously at a single time step, making it difficult to capture the continuous phase and underlying temporal context of irregular waves [15]. By incorporating LSTM structures into the deep reinforcement learning framework, the network can process continuous sequences of historical observations. This sequence learning capability helps the network understand continuous wave patterns over time. As a result, the PPO agent can maintain stable PTO damping control even in unpredictable wave conditions.

III. Technical Approach and Methods

A. System Overview: The RL-WEC Control Loop

The proposed system maximizes the energy extraction of a WEC in irregular ocean waves using a Deep Reinforcement Learning (DRL) As shown in Figure 1, the architecture operates as a closed-loop control system. JONSWAP spectrum-based irregular waves drive the WEC simulator, which is strictly limited to a $\pm 2.5m$ displacement to ensure hardware safety. Before reaching the controller, an Observation Processing module filters the wave data and builds an observation history. To train the model on long time-series data without losing context, a custom carry-over logic transfers internal variables and observation histories between partitioned training episodes.

The main controller is a PPO-LSTM agent. The LSTM architecture helps the agent understand the time-dependent patterns of the waves. The agent outputs a damping control action, and its performance is scored by an Energy Efficiency reward function. This function compares the actual extracted power against the available wave energy flux, guiding the agent to find the most efficient and safe control strategy.

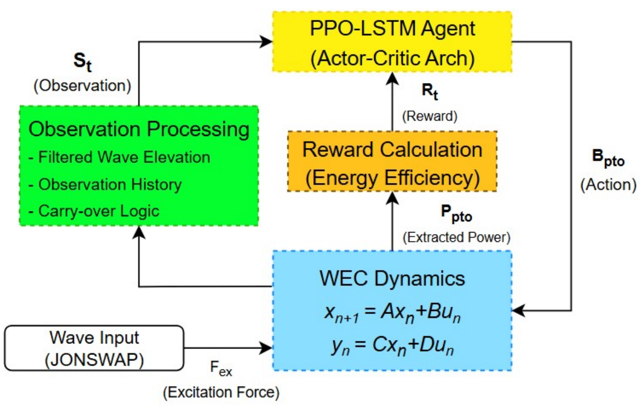

Fig. 1. Block diagram of the proposed PPO-LSTM control system for the Wave Energy Converter

B. Problem Formulation

To optimize the PTO damping control using a reinforcement learning approach, the interaction between the WEC and the irregular ocean environment is formulated as a Markov Decision Process (MDP). The environment simulates the WEC dynamics subjected to JONSWAP irregular waves, continuously providing states and evaluating actions based on physical boundaries.

Instead of depending on instantaneous values, the agent observes a state vector s_t that incorporates both current measurements and historical data to capture the wave's temporal dynamics. The observation space includes:

$$\text{state } (s_t) = [z, \dot{z}, F_{exf}, F_{exp}, B_{\text{pto},t-1}, \eta, \hat{\eta}]$$

$$\text{action } (a_t) = B_{\text{pto}}$$

Where $z, \dot{z}$ are the relative displacement and relative velocity between two bodies, F_{exf}, F_{exp} are the float and plate excitation forces, $B_{\text{pto},t-1}$ is the previous damping coefficient prior to one step, and $\eta, \hat{\eta}$ are the current wave elevation and filtered current wave elevation, respectively. Inputs to the WEC plant are float excitation force, plate excitation force, and PTO force.

The action a_t chosen by the agent corresponds to the PTO damping coefficient (B_{pto}). To maintain operational stability, the continuous action space is bounded within a range of $[0 \text{ to } 1.2 \times 10^7] N \cdot s/m$.

To enable the sequence-learning capability of the LSTM, a history window of 100 time steps is stacked. Furthermore, to maintain context across segmented training episodes, the internal variables of the reward function and the observation history are carried over to subsequent segments. This ensures continuous sequential awareness without breaking the physical timeline.

A strict physical displacement constraint is imposed on the WEC floats since they cannot oscillate indefinitely. The vertical displacement (z) is bounded to exactly $\pm 2.5m$. If the agent attempts a control strategy that drives the float beyond this $\pm 2.5m$ limit, the action is penalized, preventing the policy from exploiting unrealistic mechanical behaviors.

C. Physics-based Reward Formulation

The objective of the WEC control system is to maximize energy extraction. However, defining the reward simply as the raw extracted energy is suboptimal in an irregular wave environment, as the absolute energy potential heavily fluctuates depending on the varying scales of the incoming waves. To normalize the learning signal across diverse wave states, a physics-based Energy Efficiency (EE) reward function is proposed.

Instead of accumulating the extracted energy, the reward formulation focuses on the comparison of average power. The Energy Efficiency is defined as the ratio of the time-averaged extracted power from the PTO system to the available wave energy flux over a given evaluation window:

$$R_{EE} = \frac{\bar{P}_{\text{pto}}}{\bar{P}_{\text{wave}}} = \frac{\text{Average Extracted Power}}{\text{Available Wave Energy Flux}}$$

This normalization ensures the PPO-LSTM agent learns a generalized optimal control policy that maximizes extraction efficiency regardless of the wave magnitude.

In addition, to enforce the physical safety of the WEC, the final reward function R_t integrates a penalty mechanism. If the agent's action causes the relative displacement to exceed the strict $\pm 2.5m$ mechanical limit, a negative reward is applied:

$$R_t = \begin{cases} R_{EE}, & \text{if } |z| \leq 2.5 \\ \text{Penalty}, & \text{if } |z| > 2.5 \end{cases}$$

By combining the normalized power efficiency with the boundary penalty, the reward function prevents the agent from finding reward hacking strategies, which guides it toward a physically viable and efficient control policy.

D. PPO-LSTM Agent Architecture

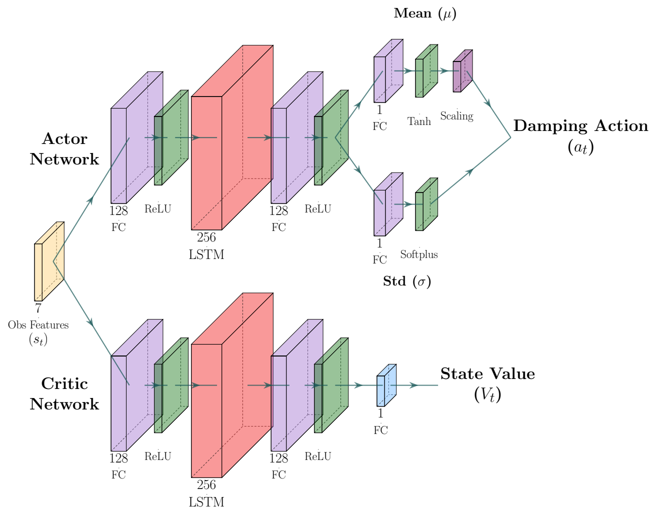

Fig. 2. PPO-LSTM Actor-Critic Architecture

To achieve continuous and stable control over the PTO damping coefficient, the Proximal Policy Optimization (PPO) algorithm is used. While standard Multi-Layer Perceptron (MLP) networks evaluate instantaneous observations, the irregular dynamics of JONSWAP waves require temporal awareness. To overcome this, a unified PPO-LSTM architecture is designed, as illustrated in Figure 2. With this structure, the agent can capture temporal dependencies and optimize energy extraction. The data processing flows through the network in three progressive stages. First, the shared input layer, consisting of the 7 observation features (s_t), is fed independently into the Actor and Critic networks. In both branches, the raw observation features are processed through fully-connected (FC) layers with ReLU activation to extract high-dimensional, temporally invariant feature representations. Next, the extracted feature sequences are passed into the LSTM layer (256 units). The recurrent structure of the LSTM processes the sequences sequentially, capturing the temporal context of the wave dynamics and the corresponding WEC states. This ensures the agent understands how the environment gradually changes over the given history window. Finally, the outputs of the LSTM are mapped to their respective final values via the back-end FC layers. In the Actor network, the architecture branches into two paths to compute the mean (μ) and standard deviation (σ) of the policy distribution. The mean branch utilizes a Tanh function with a custom Scaling layer, while the standard deviation branch employs a Softplus function. These parameters collectively determine the final sampled continuous damping action ($a_t = B_{\text{pto}}$). Concurrently, the Critic network utilizes a single linear FC layer to estimate the state value (V_t). It evaluates the overall quality of the observation sequence to guide the Actor's policy updates.

IV. Experimental Setup

This section details the simulation environment, the wave dataset configuration, the training strategies, and the hyperparameter settings used to evaluate the proposed control framework.

A. Simulation Platform and WEC Model

The entire closed-loop control system, including the WEC hydrodynamics, mechanical constraints, and the DRL agent, is implemented within the MATLAB and Simulink (R2025a) environment. The WEC plant is modeled after Reference Model 3 (RM3), a standard two-body floating point absorber. To achieve high-fidelity control and capture precise physical interactions, the simulation operates at a sampling frequency of $10Hz$, corresponding to a fixed time step of $\Delta t = 0.1s$. To manage the heavy computational demands of simulating complex marine environments over long time horizons, training is accelerated via High-Performance Computing (HPC) cluster managed by a Slurm scheduler, utilizing parallel workers across dedicated CPU and GPU resources.

B. Wave Environment Configuration (Sea State 5)

To test the PPO-LSTM agent in realistic conditions, an irregular wave dataset is generated using the JONSWAP spectrum. The simulation focuses on Sea State 5, which is a common operating environment for wave energy converters. The specific wave parameters are listed in Table I. The dataset contains 1,000 unique wave time-series, each lasting 1,200 seconds. The data is split into 80% for training and 20% for validation.

TABLE I
PARAMETERS FOR JONSWAP WAVE SPECTRUM (SEA STATE 5)

Parameter	Value	Unit
Significant Wave Height (H_s)	3.66	m
Peak Period (Tp)	9.7	s
Rho (ρ)	1000	kg/m^3
Gravitational Acceleration (g)	9.81	m/s^2
Peak Enhancement Factor (γ)	3.3	-
Simulation Duration	1,200	s
Time Step (Δt)	0.1	s

C. Training Strategy and Carry-Over Logic

At a $10Hz$ sampling rate, a full 1,200-second wave sequence generates 12,000 time steps. Training on this entire sequence at once causes unstable gradient updates. To optimize training Efficiency, each wave is partitioned into shorter sub-episodes of 200 steps (20 seconds). This length was empirically selected to balance gradient stability and the effective learning horizon determined by the discount factor. The agent was trained for a total of 48,000 sub-episodes.

To prevent physical discontinuities at the segment boundaries, a variable carry-over mechanism is introduced. Instead of transferring complex LSTM hidden states, the environment resets them but passes reward and observation variables to the next sub-episodes. Specifically, the final step of the state observation history and the exact zero-crossing detection states are saved and used to initialize the subsequent segment. This mechanism ensures the agent perceives the partitioned episodes as a single, continuous physical timeline, which is essential for accurate reward assignment and stable learning in irregular waves.

D. Hyperparameter Configuration

To guarantee the algorithmic validity and reproducibility of the trained control policy, the hyperparameters configured within the MATLAB Reinforcement Learning Toolbox are defined. These parameters dictate the optimization landscape and convergence properties of the PPO-LSTM architecture, as summarized in Table II.

TABLE II
HYPERPARAMETERS FOR PPO-LSTM TRAINING

Parameter	Value
Experience Horizon	400
Learning Frequency	400
Mini Batch Size	100
Discount Factor	0.99
Number of Epochs	5
Entropy Loss Weight	0.001
Actor Learning Rate	1×10^{-3}
Actor Gradient Threshold	5
Critic Learning Rate	1×10^{-3}
Critic Gradient Threshold	5
Sample Time	0.1

V. RESULTS

A. Benchmark Baselines and Evaluation Metrics

To evaluate the performance the performance of the proposed PPO-LSTM controller, two distinct baseline control strategies from traditional ocean engineering are established for comparative analysis. For a fair evaluation, all simulations, including the baselines and the proposed learning-based agent, are executed with a time-step of 0.1s. Furthermore, all control frameworks are restricted to damping-only control, ensuring that performance gains are driven by behavioral control logic rather than hardware or stiffness variations.

- **Fixed-Damping Control (Baseline 1)**: The conventional passive control strategy where the damping coefficient (B_{pto}) remains constant throughout the test duration. As illustrated in Figure 3, an exhaustive brute-force grid search was conducted over the range of 0 to $12MN \cdot s/m$ with a step size of $0.3MN \cdot s/m$. The optimal damping coefficient was determined to be $5.4MN \cdots /m$, yielding a maximum mean cumulative energy extraction of $3.11 \times 10^8 J$, which serves as the primary benchmark.

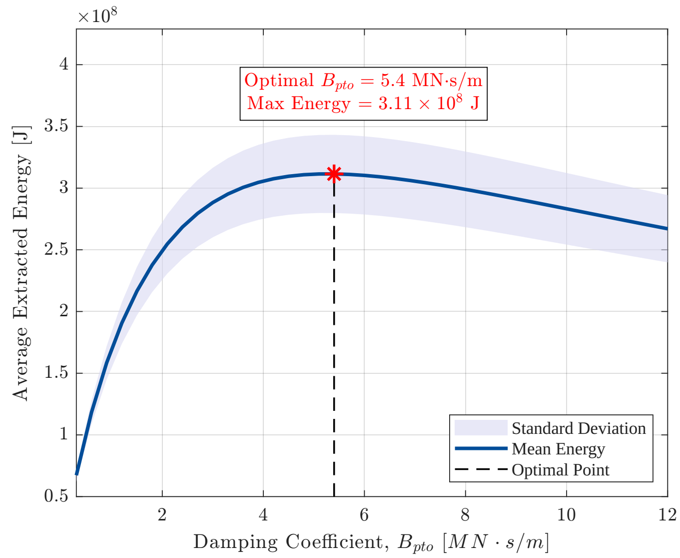

Fig. 3. Mean cumulative absorbed energy across 200 validation wave scenarios as a function of the PTO damping coefficient (B_{pto}). The optimal baseline performance is achieved at $B_{\text{pto}} = 5.4$ MN·s/m, yielding a maximum energy of 3.11×10^8 J.

- **Instantaneous Frequency (IF) Control (Baseline 2)**: A state-of-the-art non-learning active control method that adaptively adjusts the damping command based on the real-time wave frequency. For a fair comparison, we disabled the stiffness control in the IF algorithm so that it relies on damping control, matching the conditions of the other models.

The control efficiency and power extraction performance are quantified using the following two metrics, evaluated over a validation set consisting of 200 distinct irregular wave scenarios:

1) **Mean Cumulative Absorbed Energy:** The average of the total electrical energy extracted at the end of each evaluation episode across the 200 test wave profiles.

2) **Capture Width Ratio (CWR):** A non-dimensional efficiency metric commonly used in wave energy conversion to evaluate how effectively the device captures energy relative to the incoming wave power. It is defined as:

$$\text{CWR} = \frac{P_{absorbed}}{P_{incident} \cdot B}$$

where $P_{absorbed}$ represents the average power captured by the WEC, $P_{incident}$ is the wave energy flux per unit crest length, and $B = 20$ denotes the width of the device.

B. Training Convergence and Reward Function Analysis

This section analyzes how the reward function design affects training stability. For this analysis, a basic Multi-Layer Perceptron (MLP) architecture is used. The observation space is simplified to include relative displacement (z), relative velocity

($\dot{z}$), wave excitation forces (F_{exf}, F_{exp}), previous damping value, and current wave elevation.

Although the model was trained for a total of 48,000 episodes, Fig. 4 plots only the first 10,000 episodes. The training curves stabilized completely after this point, showing no significant changes in the remaining episodes.

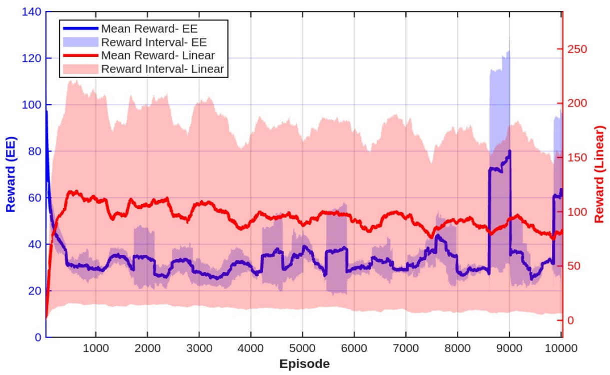

Fig. 4. Training convergence curves for the MLP architecture over the first 10,000 episodes. The proposed EE reward (blue) shows much better stability and lower variance than the alternative linear reward (red).

In Fig. 4, the proposed physics-based EE reward (blue) is compared against an alternative linear reward function (red). Please note that this linear reward is simply a different reward design used for comparison, not the fixed-damping control baseline described in Section 5.A. Specifically, the linear reward is calculated by scaling the raw extracted power by 3×10^{-6} and subtracting the same boundary penalty term used in the EE reward.

The linear reward shows a very wide reward interval. This is because the linear formulation relies on raw extracted power, which changes wildly depending on the size of the waves. In contrast, the proposed EE reward maintains a very tight and consistent interval. By dividing the captured power by the incoming wave energy flux, the EE reward removes the effect of wave scales and gives the agent steady feedback.

The sharp drop in the blue curve before episode 500 shows the agent quickly learning to avoid basic physical penalties and boundaries. After this initial phase, the EE reward stays within a stable range. The continuous minor fluctuations in the curve are caused by the random nature of the JONSWAP wave spectrum. Because each episode mixes different wave sizes randomly, some episodes are naturally harder to control than others. This leads to small ups and downs in the learning curve.

C. Control Performance Comparison

This section analyzes the power extraction efficiency of the proposed PPO-LSTM controller compared to the benchmark strategies and the theoretical upper limit across 200 irregular wave scenarios. The exact values of the mean cumulative absorbed energy and the corresponding percentage improvements over the Fixed-Damping baseline are summarized in Table III.

TABLE III
PERFORMANCE SUMMARY OF CONTROL METHODS

Control Method	**Mean Energy** (10^8 **J**)	**Improvement (%)**
Theoretical Max	6.30	-
Fixed-Damping (Baseline)	3.11	Base
Instantaneous Frequency (IF)	3.37	+8.36%
Proposed PPO-LSTM	**3.38**	**+8.68%**

As shown in Table III, both active control methods extract more energy than the passive Fixed-Damping baseline (3.11×10^8J). Because the passive baseline relies on a single constant damping coefficient, its performance is limited in fluctuating, stochastic wave environments, which underscores the necessity of active control.

The proposed PPO-LSTM controller achieves an average energy extraction of 3.38×10^8J (an 8.68% increase over the baseline), which is comparable to the state-of-the-art IF active control method (3.37×10^8J, 8.36% improvement). This result indicates that the learning-based framework can closely match the performance of advanced ocean engineering control strategies. Notably, the PPO-LSTM agent achieves this near state-of-the-art efficiency without requiring real-time wave frequency calculations during operation, presenting a distinct practical advantage.

To further evaluate the statistical robustness of the controllers across all 200 test scenarios, the distribution of the Capture Width Ratio (CWR) is presented in Fig. 5.

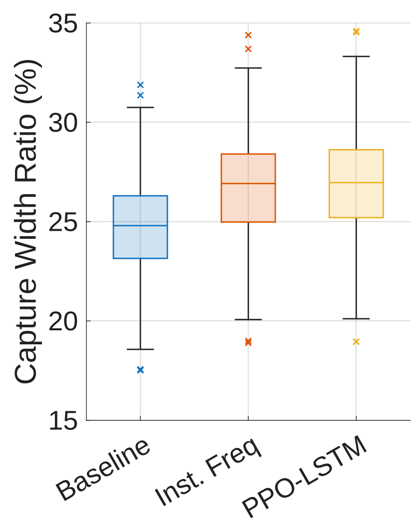

Fig. 5. Comparison of Capture Width Ratio (CWR) distributions across 200 validation irregular wave scenarios.

As illustrated in Fig. 5, the CWR distribution of the PPO-LSTM controller closely aligns with that of the advanced IF method. While there are no prominent statistical deviations between the two active control distributions, the overall efficiency profiles of both methods show a upward shift compared to the passive baseline.

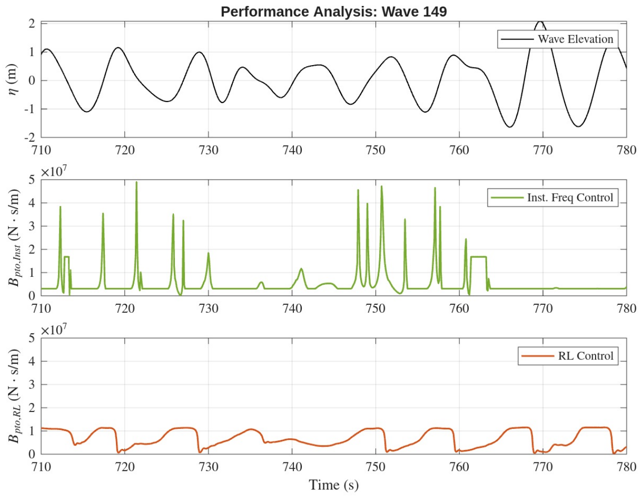

Fig. 6. Time-series control behavior under irregular wave scenario 149: (a) wave elevation, (b) damping coefficient command of the IF controller, and (c) damping coefficient command of the proposed PPO-LSTM controller.

The fact that the learning-based model replicates the statistical efficiency distribution of a state-of-the-art engineering controller is highly meaningful. It demonstrates that even with a straightforward neural network structure, the agent effectively captures the sequential features of the irregular waves. The proposed model provides reliable and practical alternative that yields robust efficiency across diverse JONSWAP wave profiles.

D. Dynamic Control Behavior Analysis

To understand how the controllers operate in real-time, a time-series analysis is conducted using a representative irregular wave segment (Wave 149). Fig. 6 shows the wave elevation, the damping command of the proposed PPO-LSTM model, and the damping command of the IF method over the same time period.

As shown in Fig. 6(c), the proposed PPO-LSTM controller adjusts the damping coefficient smoothly and continuously. It keeps the control values low (under $1.1 \times 10^7 N \cdot s/m$) and avoids sudden jumps. Because the LSTM layer remembers past data, the network can follow wave trends and make stable, proactive adjustments.

In contrast, Fig. 6(b) shows that the IF method produces sharp and sudden spikes. To react to instant frequency changes, it applies rapid impulses that reach much higher values, up to $5 \times 10^7 N \cdot s/m$.

This comparison shows a major practical advantage of the proposed model. The PPO-LSTM agent achieves similar energy efficiency to the IF method but maintains a much smoother and lower control profile. This significantly reduces physical stress on the mechanical parts of the device.

VI. Summary and Future Work

This paper presented a reinforcement learning-based control strategy to increase the energy harvesting efficiency of wave energy converters (WECs). The proposed agent learns to adjust the PTO damping coefficient at the exact timing under irregular wave conditions. Using a straightforward neural network structure, the model achieved a mean cumulative energy extraction of 3.38×10^8J and a Capture Width Ratio (CWR) of 27.1%. These results are comparable to state-of-the-art active control methods in ocean engineering. This highlights the promising capability of learning-based frameworks to provide adaptive and efficient control without requiring online frequency calculations.

For future research, three main directions will be explored to build upon these findings:

- **Explore alternative architectures and algorithms:** Different neural network structures, such as Transformers, and alternative reinforcement learning algorithms will be investigated to unlock new potential for system efficiency.
- **Validate agent robustness in complex sea states:** The control policy will be tested against more challenging environmental conditions, including mixed wave profiles and extreme sea states, to ensure operational reliability.
- **Develop a hybrid control framework:** We plan to combine the existing instantaneous frequency control logic with the reinforcement learning policy. This hybrid approach can help guide the learning process and further maximize power capture efficiency.

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